

Embedded DTMC in a CTMC

- ▶ Consider a CTMC over state space $\mathcal{S}$.
- ▶ Let $Y_n, n \geq 0$ denote the sequence of times spent in successive states of the CTMC
- ▶ Define T_n to be the jump times of the CTMC, i.e., the times of successive state transitions.
- ▶ Then $T_n = \sum_{k=1}^n Y_k$.
- ▶ Define $X_n = X(T_n)$ for $n \geq 0$. X_n is the DTMC embedded in the CTMC.
- ▶ The corresponding TPM has $p_{ij} = \frac{q_{ij}}{|q_{ii}|}$.
- ▶ $\{X_n, \}$ is such that there are no one step transitions from a state to itself, i.e., $p_{ii} = 0$.

Transience and Recurrence

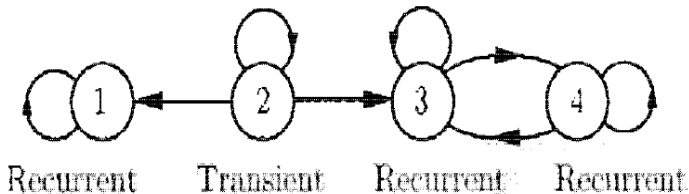
Classification of states

- ▶ Consider a Markov process with state space $\mathcal{S}$
- ▶ We say that j is accessible from i if $p_{ij}^n > 0$ for some n .
- ▶ This is denoted by $i \rightarrow j$.
- ▶ if $i \rightarrow j$ and $j \rightarrow i$ then we say that i and j communicate. This is denoted by $i \leftrightarrow j$.

A chain is said to be irreducible if $i \leftrightarrow j$ for all $i, j \in \mathcal{S}$.

Recurrent and Transient states

- ▶ We say that a state i is recurrent if $F_{ii} = P(\text{ ever returning to } i \text{ having started in } i) = 1$.
- ▶ F_{ii} is not easy to calculate. (We will see this after quiz)
- ▶ If a state is not recurrent, it is transient.
- ▶ For a transient state i , $F_{ii} < 1$.
- ▶ If $i \leftrightarrow j$ and i is recurrent, then j is recurrent.



First recurrence probabilities

- ▶ Define: $F_{ii} = \sum_{n=1}^{\infty} f_{ii}^n$.
- ▶ f_{ii}^n : probability of starting in i and returning to state i for the first time exactly after n steps.
- ▶ $f_{ii}^n := P(X_n = i, X_k \neq i \text{ for } 1 \leq k \leq n-1 | X_0 = i)$. ($f_{ii}^0 = 0$).
- ▶ F_{ii} has the interpretation of the probability of ever returning to state i .
- ▶ If $F_{ii} = p < 1$, then there is a finite probability $1 - p$ with which you may not return to state i .
- ▶ If $F_{ii} = 1$, then from i you can certainly return to i .
- ▶ For any $i \in \mathcal{M}$, the first return time T_{ii} has the probability mass function $\{f_{ii}^n, n \geq 0\}$.

Mean passage and recurrence times

- ▶ Let μ_{ii} be the mean recurrence time at state i , i.e. $E[T_{ii}]$.
- ▶ $\mu_{ii} = \sum_{n=1}^{\infty} n f_{ii}^n$
- ▶ All the above definitions have an equivalent counterpart in a CTMC.
- ▶ For eg: f_{ii}^t has a natural interpretation. We wont go further into this.

Transient and recurrent states

- ▶ Suppose for a state i we have $F_{ii} = 1$. Then we say that state i is recurrent.
- ▶ Once in i , you are certain to come back to i .
- ▶ If $\mu_{ii} = \infty$, it is called null recurrent. The chain is bound to return to state i , but possibly after an infinite time.
- ▶ $\mu_{ii} < \infty$, it is called positive recurrent.
- ▶ If all states of the Markov chain are (null /positive) recurrent, it is called as a (null /positive) recurrent Markov chain.
- ▶ Null recurrence is possible in infinite state space models.
- ▶ State i is transient if $F_{ii} < 1$.
- ▶ You may not return back to i with a finite probability.

Transient and recurrent states

- ▶ Consider a DTMC and consider $X_0 = i$.
- ▶ Let us count the number of times the chain is in state i .
- ▶ Let I_n denote an indicator variable which is 1 if $X_n = i$ and 0 if $X_n \neq i$.
- ▶ $\sum_{n=1}^{\infty} I_n$ counts the number of times state i was visited.
- ▶ $E[I_n] = P(X_n = i | X_0 = i) = p_{ii}^{(n)}$.
- ▶ The mean total number of visits to state i is given by $\sum_{n=1}^{\infty} p_{ii}^n$
- ▶ Convergence or divergence of this sum also defines transient or recurrent states.

Recurrent criteria

- ▶ The mean total number of visits to state i is given by $\sum_{n=1}^{\infty} p_{ii}^n$
- ▶ Suppose the chain visits state i only exactly n times.
- ▶ The $P(\text{exactly } n \text{ visits to } i) = F_{ii}^n(1 - F_{ii})$.
- ▶ For a recurrent state, $F_{ii} = 1$. Hence $P(\text{exactly } n \text{ visits to } i) = 0$.
- ▶ $P(\text{exactly infinite visits to } i) = 1$.
- ▶ Mean total number of visits is also infinite and hence $\sum_{n=1}^{\infty} p_{ii}^n$ diverges.

Transient state criteria

- ▶ The mean total number of visits to state i is given by $\sum_{n=1}^{\infty} p_{ii}^n$
- ▶ Suppose the chain visits state i only exactly n times.
- ▶ The $P(\text{exactly } n \text{ visits to } i) = F_{ii}^n(1 - F_{ii})$.
- ▶ For transient state i , $F_{ii} < 1$.
- ▶ The $P(\text{exactly } n \text{ visits to } i) = F_{ii}^n(1 - F_{ii})$. Compare this with geometric random variable.
- ▶ Mean total number of visits to state i is $\frac{F_{ii}}{1-F_{ii}}$ which is finite.
- ▶ Hence for transient state $\sum_{n=1}^{\infty} p_{ii}^n$ must converge.

Classification of states

- ▶ Consider a Markov process with state space $\mathcal{S}$
- ▶ We say that j is accessible from i if $p_{ij}^n > 0$ for some n .
- ▶ For a CTMC, the condition is $p_{ij}(t) > 0$ for some t .
- ▶ This is denoted by $i \rightarrow j$.
- ▶ if $i \rightarrow j$ and $j \rightarrow i$ then we say that i and j communicate.
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A chain is said to be irreducible if $i \leftrightarrow j$ for all $i, j \in \mathcal{S}$.

Communicating class

- ▶ A set $\mathcal{C} \subset \mathcal{S}$ is called a communicating class if all states in $\mathcal{C}$ communicate with each other
- ▶ These states can reach to states outside the class (hence the class is not closed)
- ▶ The communicating class is closed if states from the class cannot reach outside $\mathcal{C}$.
- ▶ The above is true for DTMC as well as CTMC.

Communicating class is an equivalence relation

- ▶ A binary relation R on $\mathcal{S}$ is a subset of the product space $\mathcal{S} \times \mathcal{S}$.
- ▶ Equivalence relation is a binary relation which is symmetric, reflexive and transitive.
- ▶ $i \leftrightarrow i$ (Reflexivity)
- ▶ If $i \leftrightarrow j$ then $j \leftrightarrow i$ (Symmetry)
- ▶ If $i \leftrightarrow j$ and $j \leftrightarrow k$ then $i \leftrightarrow k$. (Transitivity)

Periodicity

- ▶ Periodicity is a class property
- ▶ Period d_i of state i is defined as $d_i = \gcd\{m : p_{ii}^{(m)} > 0\}$
- ▶ If $d_i = 1$ for state i , it is called aperiodic else its called periodic.
- ▶ For a CTMC, what is its periodicity ?
- ▶ A CTMC is generally considered aperiodic.
- ▶ What about the Embedded DTMC ? Can it be periodic ?

Recurrence and Transience as class property

- ▶ Recall the definition of closed communicating class.
- ▶ There is a positive probability with which you can go from one state to another within the class.
- ▶ All states in such a closed communicating class are either transient or recurrent.